



# Large Intestine — MCQ Practice

Anatomy GIT Block · MEDucation by Mattin Majid

## HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

**4 Easy · 10 Medium · 6 Hard** · suggested time ≈ 25 minutes.

## Q Questions

**1. In which position does the tip of the appendix most commonly lie?**

EASY

- A. Pelvic
- B. Retrocecal / retrocolic
- C. Subcecal
- D. Pre-ileal
- E. Paracolic

**2. Which part of the large intestine is the longest and most mobile?**

EASY

- A. Ascending colon
- B. Descending colon
- C. Sigmoid colon
- D. Transverse colon
- E. Rectum

**3. Which landmark divides the anal canal into an upper two-thirds and a lower one-third?**

EASY

- A. Pectinate line
- B. Anorectal junction
- C. Anal verge
- D. White line of Hilton
- E. Rectal ampulla

**4. Which artery supplies the cecum?**

EASY

- A. Right colic artery
- B. Middle colic artery
- C. Ileocolic artery (branch of the SMA)
- D. Left colic artery
- E. Superior rectal artery

**5. A 20-year-old has central abdominal pain around the umbilicus that later moves to the right iliac fossa. Why is the early pain felt around the umbilicus?**

MEDIUM

- A. The appendix lies directly behind the umbilicus
- B. The inflamed appendix irritates the femoral nerve
- C. The pain spreads along the ileocolic artery
- D. The appendix is supplied by the pudendal nerve
- E. The appendix and the umbilical region share the T10 spinal segment (referred pain)

**6. During an appendicectomy the appendix cannot be seen. Which feature of the cecum should the surgeon trace to find its base?**

MEDIUM

- A. The three taeniae coli, which converge at the base of the appendix
- B. The ileocecal valve
- C. The right colic artery
- D. The appendices epiploicae
- E. The hepatic flexure

**7. At laparotomy for intestinal obstruction, the surgeon finds the cecum distended. What does this tell him?**

MEDIUM

- A. The obstruction is in the jejunum
- B. The obstruction is in the duodenum
- C. The obstruction is in the large intestine
- D. The ileocecal valve is incompetent
- E. There is no mechanical obstruction

**8. An inflamed appendix passes upward and medially in front of the terminal ileum, with its tip pointing toward the spleen. Why is this position regarded as the most dangerous?**

MEDIUM

- A. It lies against the femoral nerve
- B. Inflammation spreads into the general peritoneal cavity
- C. It is the least common position, so it is often missed
- D. It erodes the ileocolic artery
- E. It drains to the inguinal nodes

**9. In a male, the rectum lies behind the urinary bladder. Which peritoneal pouch separates them?**

MEDIUM

- A. Uterovesical pouch
- B. Rectouterine pouch
- C. Paracolic gutter
- D. Rectovesical pouch
- E. Retrocecal recess

**10. A 45-year-old man has a very painful swelling at the anal margin, below the pectinate line. Which vein is dilated?**

MEDIUM

- A. Superior rectal vein
- B. Middle colic vein
- C. Inferior mesenteric vein
- D. Left gastric vein
- E. Inferior rectal vein

**11. A patient with portal obstruction has painless bright-red bleeding from swellings above the pectinate line. Which veins are dilated?**

MEDIUM

- A. Tributaries of the superior rectal vein
- B. Tributaries of the inferior rectal vein
- C. Tributaries of the middle colic vein
- D. The internal pudendal vein
- E. The ileocolic vein

**12. A squamous carcinoma arises in the anal canal below the pectinate line. Which lymph nodes are most likely involved first?**

MEDIUM

- A. Internal iliac nodes
- B. Inferior mesenteric nodes
- C. Superficial inguinal nodes (horizontal group)
- D. Superior mesenteric nodes
- E. Pararectal nodes

**13. An elderly patient develops sudden distension and obstruction because a loop of large bowel has twisted around its own mesentery. Which part is most likely involved, and why?**

MEDIUM

- A. Ascending colon, because it is retroperitoneal
- B. Cecum, because it has the highest pressure
- C. Descending colon, because it lies in the left paracolic gutter
- D. Sigmoid colon, because its mesentery makes it extremely mobile
- E. Rectum, because it has five curvatures

**14. A tumour arises in the distal third of the transverse colon. To which main node group will its lymph ultimately drain?**

MEDIUM

- A. Superior mesenteric nodes
- B. Inferior mesenteric nodes
- C. Internal iliac nodes
- D. Celiac nodes
- E. Superficial inguinal nodes

**15. A patient with carcinoma of the upper rectum is found to have liver metastases. By which route did the tumour cells most likely reach the liver?**

HARD

- A. Middle rectal vein → internal iliac vein → IVC
- B. Inferior rectal vein → internal pudendal vein
- C. Lymphatics → superficial inguinal nodes
- D. Direct spread through the pouch of Douglas
- E. Superior rectal vein → inferior mesenteric vein → portal vein

**16. A man with advanced rectal carcinoma develops pain radiating down the back of his leg. Which direction of tumour invasion explains this?**

HARD

- A. Posterior invasion of the sacral plexus
- B. Anterior invasion of the prostate
- C. Lateral invasion of the ureter
- D. Upward spread to the sigmoid mesentery
- E. Invasion of the ischiorectal fossa fat

**17. A lesion shows these features: ectodermal origin, stratified squamous epithelium, supply from the inferior rectal artery, somatic innervation. Which other feature belongs to the same region?**

HARD

- A. Venous drainage to the portal system via the superior rectal vein
- B. Lymph drainage to the internal iliac nodes
- C. Painless haemorrhoids
- D. Lymph drainage to the superficial inguinal nodes
- E. Autonomic innervation via the inferior hypogastric plexus

**18. An inflamed retrocecal appendix lies against the posterior relations of the cecum. Which of the following is NOT a posterior relation of the cecum?**

HARD

- A. Right psoas major and iliacus
- B. Femoral nerve
- C. Greater omentum
- D. Lateral cutaneous nerve of the thigh
- E. Right gonadal vessels

**19. A surgeon ligates the inferior mesenteric artery at its origin. Which segments lose their main named arterial supply?**

**HARD**

- A.** Cecum and ascending colon
- B.** Proximal two-thirds of the transverse colon only
- C.** Lower anal canal only
- D.** Appendix and cecum
- E.** Descending colon, sigmoid colon and upper rectum (and the left-colic supply to the transverse colon)

**20. Which statement correctly distinguishes the two anal sphincters?**

**HARD**

- A.** Both are voluntary and made of skeletal muscle
- B.** Internal: involuntary thickened circular muscle around the upper two-thirds; external: voluntary, with deep, superficial and subcutaneous parts along the whole canal
- C.** Internal: voluntary with three parts; external: involuntary circular muscle
- D.** The external sphincter surrounds only the lower third and has no parts
- E.** The internal sphincter lies outside the external sphincter

## ✓ Answer key

| Q  | Answer   | Difficulty | Why                                                                                                                                                                                                                                                                         |
|----|----------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | <b>B</b> | EASY       | The retrocecal/retrocolic position is the commonest (65.28%). Pelvic is second (31.01%).                                                                                                                                                                                    |
| 2  | <b>D</b> | EASY       | The transverse colon is about 45 cm (18 in) long and is the longest and most mobile part. Its mesentery is the transverse mesocolon.                                                                                                                                        |
| 3  | <b>A</b> | EASY       | The pectinate line divides the visceral upper two-thirds (intestine) from the somatic lower one-third (perineum).                                                                                                                                                           |
| 4  | <b>C</b> | EASY       | The cecum is supplied by the ileocolic artery from the superior mesenteric artery. Its veins drain to the SMV and its lymph to the superior mesenteric nodes.                                                                                                               |
| 5  | <b>E</b> | MEDIUM     | Early appendicitis pain is referred to the umbilical region because both share the T10 segment. Later, maximum tenderness is at McBurney's point.                                                                                                                           |
| 6  | <b>A</b> | MEDIUM     | All three taeniae coli converge at the base of the appendix, a guide the lecture highlights for appendectomy.                                                                                                                                                               |
| 7  | <b>C</b> | MEDIUM     | The cecum is the surgeon's reference point: distended cecum means the obstruction is in the large intestine; an empty cecum points to the small intestine.                                                                                                                  |
| 8  | <b>B</b> | MEDIUM     | This is the pre-ileal position (1%). The lecture calls it the most dangerous because inflammation spreads into the general peritoneal cavity.                                                                                                                               |
| 9  | <b>D</b> | MEDIUM     | In the male, the rectovesical pouch lies between the rectum and bladder. The lecture notes these pouches help limit the spread of infection.                                                                                                                                |
| 10 | <b>E</b> | MEDIUM     | External (false) piles are dilated tributaries of the inferior rectal vein below the pectinate line. They are painful because the inferior rectal nerves are somatic.                                                                                                       |
| 11 | <b>A</b> | MEDIUM     | Internal (true) piles are dilated tributaries of the superior rectal vein above the pectinate line. They are painless because that area has autonomic innervation.                                                                                                          |
| 12 | <b>C</b> | MEDIUM     | The lower anal canal drains to the horizontal group of superficial inguinal nodes. The upper part drains to the internal iliac nodes.                                                                                                                                       |
| 13 | <b>D</b> | MEDIUM     | Sigmoid volvulus: the extreme mobility of the sigmoid colon on its mesentery lets it rotate. The same mobility makes it the usual colostomy site.                                                                                                                           |
| 14 | <b>B</b> | MEDIUM     | The proximal two-thirds of the transverse colon drain via colic nodes to the superior mesenteric nodes; the distal third drains to the inferior mesenteric nodes.                                                                                                           |
| 15 | <b>E</b> | HARD       | The superior rectal vein is a tributary of the portal circulation via the inferior mesenteric vein, so rectal cancer can metastasize to the liver.                                                                                                                          |
| 16 | <b>A</b> | HARD       | Posterior penetration of the rectal wall can invade the sacral plexus and cause sciatica. Lateral spread reaches the ureter; anterior spread reaches the bladder, prostate, seminal vesicles, vagina or uterus.                                                             |
| 17 | <b>D</b> | HARD       | All the listed features belong to the lower third of the anal canal (below the pectinate line), which also drains to the superficial inguinal nodes and systemic veins. The other options describe the upper two-thirds.                                                    |
| 18 | <b>C</b> | HARD       | The greater omentum is an anterior relation, with the small-intestine coils and anterior abdominal wall. Posteriorly lie psoas and iliacus, the femoral, lateral cutaneous and genitofemoral nerves, the gonadal vessels and the retrocecal recess.                         |
| 19 | <b>E</b> | HARD       | The IMA gives the left colic (descending colon, and part of the transverse), sigmoid branches and the superior rectal artery. The cecum, ascending and proximal transverse colon depend on the SMA; the lower anal canal on the inferior rectal (internal pudendal) artery. |
| 20 | <b>B</b> | HARD       | The internal sphincter is the involuntary thickened circular coat around the upper two-thirds, surrounded by the deep and superficial external parts. The voluntary external sphincter runs the whole length and has three parts.                                           |